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To the Editors of *Progress in Physics*
(Dr. Pierre Millette; Dr. Andreas Ries)

Dear Editors,

I am pleased to submit for your consideration the manuscript

"Unified Coulomb Theory for Atom and Nuclear Physics"

for publication in *Progress in Physics* as a research article. In accordance with the journal's guidelines the manuscript is submitted as a preliminary PDF; a LaTeX version prepared in the `pip-template` will be provided upon acceptance.

What the paper claims. The manuscript advances a single thesis: that atomic and nuclear structure are one and the same physics — opposite-sign charge clouds interacting by the Coulomb force alone, on non-overlapping domains that meet at free boundaries and minimise the electrostatic energy (the RealQM framework) — and that the two are related by a single operation, the interchange of electron and proton, $e^- \leftrightarrow p$. Because the kinetic energy is written with the same form for both species, this swap is an *exact symmetry* of the model rather than an analogy. Under it the ladder of ordinary matter (atom → molecule → solid) maps term by term onto the nuclear ladder (alpha → alpha-cluster → nuclear matter): closed electron shells (noble gases) correspond to closed proton shells (the alpha and the magic nuclei), covalent bonds by shared electrons to nuclear bonds by shared protons (alpha-cluster states / nuclear molecules), and condensed-matter saturation to nuclear saturation.

What is new and testable. Built from a single repeating unit — a (2 electrons + 4 protons) alpha — the computed alpha-conjugate binding-energy ladder reproduces the experimental binding energies of ^{4}He, ^{12}C, ^{16}O, …, ^{40}Ca to a uniform ∼107%, with near-constant binding per nucleon, using no strong or weak force and no fitted parameter beyond a single energy scale fixed on the deuteron. The paper also argues that alpha-clustering is *forced* (a monolithic nucleus over-binds as Z^2, so extensive, saturating binding requires localising the Coulomb glue into alpha-sized packets), and it reports a computed, free-boundary metastable alpha geometry.

A foundational corollary. Analysing the two scales that coexist in a single helium atom (an ångström electron cloud around a femtometre nucleus) leads to the conclusion that it is *size*, not mass, that is the electromagnetic variable: the same electron takes the proton's femtometre size in the nucleus and a size 10^5 times larger in the atom, at fixed charge, the difference registered entirely in the kinetic energy. Mass enters the Coulomb energetics only as an overall scale, not as a coupling in the Coulomb law, and on this reading belongs, with gravitation, to a sector separate from electromagnetism.

Why *Progress in Physics.* The work is foundational and cross-disciplinary, bridging atomic and nuclear physics from first principles and questioning a standard assumption — that nuclear binding requires a strong force. It is parameter-free and openly falsifiable, and every computation is reproducible: an open-source, browser-based (WebGPU) RealQM solver and all the simulations cited in the paper — for both atoms and nuclei — are publicly available in an interactive gallery (`https://claes542.github.io/RealMolecule/gallery.html`; source `https://github.com/Claes542/RealMolecule`). I believe this fits the journal's scope for original, independent, foundational physics.

Declarations. The manuscript is original, is not under consideration elsewhere, and has not been previously published. There are no competing interests. The mathematical model and the physical argument are my own; the numerical implementation was developed and run with AI assistance (Claude, Anthropic), as stated in the Acknowledgements. A companion paper, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement*, provides the detailed nuclear computations summarised here and is cited accordingly.

I would be glad to suggest reviewers or to provide any further material the editors may require. Thank you for your consideration.

Sincerely,

Claes Johnson
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